Koa Shen

Mechanical Engineer

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Aurora Simulation Engineering Hiring Team

Re: Summer 2027 Simulation Engineering Intern

Dear Aurora Hiring Team,

After speaking with Aurora's team about the Mechanical Design Engineering Internship, I was encouraged to apply for the Summer 2027 Simulation Engineering Intern role. I appreciated the conversations and am excited by the opportunity to contribute to the simulation systems Aurora uses to develop and validate the Aurora Driver. The role's focus on 3D geometry, physically meaningful simulation, and validating models against real-world evidence closely matches the work I have found most engaging.

At SLAC National Accelerator Laboratory, I built an open-source Python framework on Drake for kinematics and collision checking in tightly packaged X-ray beamline assemblies. I developed CAD ingestion using OpenCascade, generated collision geometry with CoACD, and built a reusable catalog of 15 motion stages. A secondary mesh triangle-triangle distance query provides an independent proximity check. Applied to a 30-DOF, 305-body assembly, millisecond-scale checks exposed at least five genuine physical pinch points and more than 10 actionable design changes. I also compared modeled geometry with the physical assembly and surfaced outdated CAD, reinforcing that simulation is only useful when its inputs and results are checked against hardware.

My independent actuator project adds hands-on C++ experience. I designed and fabricated a compact cycloidal reducer and wrote embedded PlatformIO firmware for stepper control and encoder feedback. On a single-joint bench setup, I validated position-error correction and missed-step detection. This work has given me practice connecting software behavior to measurable physical motion and debugging the full system across mechanical and electrical interfaces.

I would bring Aurora experience building simulation tools, reasoning about kinematics and 3D collision geometry, and checking models against physical systems, along with a willingness to learn the performance and validation demands of production-scale simulation. I would be excited to help develop robust tools that let engineers test the Aurora Driver safely and efficiently.

Thank you for your consideration. I would welcome the opportunity to discuss how my simulation and hardware experience could contribute to Aurora's team.

Sincerely,
Koa Shen